

Group 7 CorroCheck

CorroCheck Prototype Design Rationale

Introduction

The CorroCheck mobile application prototype was designed to provide a professional, clean, and user-friendly platform for corrosion inspection and reporting. The design focuses on simplicity, usability, and efficiency to support users performing infrastructure inspections in real-world environments.

The prototype follows a mobile-first approach because the application is intended for field use on smartphones and portable devices.

Visual Design Choices

The interface uses a blue and gray color palette to communicate trust, professionalism, safety, and technical reliability. Blue was intentionally chosen because it is commonly associated with engineering, technology, and stability, making it suitable for an industrial inspection application.

A clean and minimal design style was used throughout the prototype to reduce clutter and improve readability. Consistent typography, spacing, icons, and card layouts were applied to create a cohesive and visually balanced interface.

Navigation and User Flow

Bottom tab navigation was implemented to allow users to move easily between the main sections of the application, including Home, Upload, History, and Profile. This structure supports simple mobile navigation and improves accessibility for users.

The prototype includes two main user journeys:

1. Authentication flow: Splash Screen → Sign Up → Login → Home
2. Inspection workflow: Home → Upload → Results → History → Inspection Details

These flows were designed to ensure users can complete tasks efficiently and intuitively.

Functional Screens and Feedback States

The prototype includes the main functional screens required for the application, such as authentication screens, upload functionality, inspection results, history tracking, and profile management.

To improve usability and simulate realistic application behavior, the prototype also includes feedback states such as:

- loading indicators,
- success alerts,
- error messages,
- and empty states.

Examples include upload loading spinners, inspection success confirmations, and empty history states for users with no recorded inspections.

Conclusion

The CorroCheck prototype was designed to create a reliable and user-centered mobile experience for corrosion inspection and reporting. Through the use of clean visuals, intuitive navigation, and clear feedback states, the prototype demonstrates an efficient and professional inspection workflow while aligning with the project requirements and engineering-focused design goals.

Prototype link: <https://www.figma.com/proto/9Tr9lqcNSDtei18gyi0bn8/Corrosion-App-UI-Design-System?node-id=1-2&t=bWEaf3XfufAsTNH4-1>

The prototype was designed using the iPhone 15 Pro frame in Figma because its dimensions closely matched the screen specifications provided for the project.